MIEB: Massive Image Embedding Benchmark

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Abstract

Image and image-text embeddings underpin a wide range of vision and multimodal applications, yet their evaluation remains fragmented across narrow, task-specific protocols. We introduce ***MIEB*** *(Massive Image Embedding Benchmark), a unified and comprehensive benchmark comprising 130 tasks across 8 task categories and 38 languages. MIEB extends the MTEB [25] philosophy to vision, standardizing evaluation across retrieval, document understanding, linear probing, zero-shot classification, clustering, compositionality, vision-centric QA, and a novel* Visual STS *protocol. Visual STS diagnoses visual text understanding by rendering standard STS sentence pairs as images and computing image-image similarity. We evaluate 20 models spanning contrastive encoder families (CLIP [30], SigLIP [37], EVA-CLIP [31]) and multimodal LLM-based encoders (E5-V [16], VLM2Vec [17], Voyage-multimodal-3 [1]). No single model dominates all categories: Voyage-multimodal-3 achieves the highest overall mean (62.5), while encoder models excel on classification and clustering. We further introduce MIEB-lite, a 51-task subset derived via task-correlation analysis that reduces evaluation runtime by 82.4% with near-perfect ranking preservation (Spearman* $\rho = 0.992$*). MIEB and a public leaderboard are available at* `https://hf.co/spaces/mteb/leaderboard`*.*

1. Introduction

Image embeddings are the cornerstone of modern visual retrieval, classification, and multimodal understanding systems. From semantic image search to cross-lingual image-text retrieval, the quality of image representations determines downstream performance. Yet despite their importance, rigorous and comprehensive evaluation of these embeddings has lagged behind the standardization achieved in text-only settings.

Existing evaluation protocols for image and image-text embeddings suffer from several limitations. First, they are often *task-specific*: a model optimized for zero-shot ImageNet classification may be evaluated only on that task, while a retrieval-focused model is assessed only on standard retrieval sets. This fragmentation makes cross-model comparison unreliable and prevents practitioners from understanding which model best suits their use case. Second, important capabilities remain *under-evaluated*: visual text understanding (OCR-like abilities), interleaved image-text inputs, compositional reasoning, and multilingual support receive little systematic attention. Third, the lack of a unified, reproducible evaluation framework makes it difficult to track progress over time.

The text embedding community addressed a similar fragmentation problem with the Massive Text Embedding Benchmark (MTEB) [25], which standardized evaluation across diverse NLP tasks, languages, and domains. MTEB's single leaderboard dramatically clarified the landscape of text embedding models. We take inspiration from MTEB and extend this philosophy to vision.

We present **MIEB** (Massive Image Embedding Benchmark), a unified benchmark for image and image-text embeddings covering 130 tasks across 8 categories and 38 languages. MIEB evaluates models in a zero-shot setting (except for lightweight linear probes on frozen embeddings) to assess general-purpose representation quality. The benchmark encompasses standard vision tasks (zero-shot classification, linear probing, clustering, image-text retrieval) as well as under-explored capabilities (document understanding, compositional reasoning, vision-centric QA, and multilingual retrieval).

A central novel contribution of MIEB is **Visual STS**: a protocol that renders standard Semantic Textual Similarity (STS) sentence pairs as images and evaluates models by their image-image similarity. This allows us to directly probe whether vision encoders have internalized text understanding, bridging the gap between text and vision evaluation. We find that Visual STS scores strongly predict performance on generative MLLM benchmarks requiring OCR and chart understanding.

To make evaluation practical at scale, we introduce **MIEB-lite**, a 51-task subset derived by identifying representative tasks through task-correlation analysis using UMAP [24] and HDBSCAN [23]. MIEB-lite reduces runtime by 82.4% for large 8B models while preserving a near-perfect Spearman correlation of 0.992 with full-benchmark rankings.

Our main contributions are:

- **MIEB**: a 130-task benchmark spanning 8 categories and 38 languages with a public leaderboard for image and image-text embeddings.
- **Visual STS**: a novel evaluation protocol that renders text similarity tasks as images, diagnosing visual text understanding in encoders.
- **MIEB-lite**: an efficient 51-task subset with 82.4% runtime reduction and ρ=0.992 correlation to the full benchmark.
- Comprehensive empirical analysis of 20 models, revealing complementary strengths of encoder families and MLLM-based models, and establishing Visual STS as a proxy for MLLM encoder selection.

2. Related Work

Image Embedding Evaluation. Contrastive vision-language models like CLIP [30] and SigLIP [37] are typically evaluated on zero-shot ImageNet classification and standard image-text retrieval benchmarks (MS-COCO, Flickr30k). While the LAION CLIP benchmark [20] consolidates several such tasks, it focuses primarily on zero-shot classification and image-text retrieval, leaving document understanding, clustering, and multilingual generalization underexplored. BEIR [32] standardized text retrieval evaluation; our work extends analogous unification to vision. MMTEB [11] recently expanded MTEB to multilingual text tasks; MIEB provides the corresponding unification for image modalities.

Multimodal Language Model Embeddings. Recent work has explored using MLLMs as embedding models. VLM2Vec [17] fine-tunes Phi-3.5-Vision with contrastive learning for multi-task multimodal retrieval. E5-V [16] leverages instruction tuning to produce strong universal image-text embeddings. Voyage-multimodal-3 [1] and VISTA [39] similarly produce high-quality multi-modal embeddings. These MLLM-based approaches excel on OCR-heavy tasks where text within images must be understood semantically.

Compositionality and Fine-grained Evaluation. ARO [36], WinoGround [34], and SugarCrepe [14] probe compositional reasoning in vision-language models by testing sensitivity to attribute-relation-order modifications. MIEB incorporates these as a Compositionality category alongside standard retrieval and classification.

Document Retrieval. ColPali [12] introduced document retrieval benchmarks (VIDORE) that require understanding document page images, including charts, tables, and text. MIEB includes document understanding tasks inspired by this line of work.

Linear Probing and Representation Quality. Linear probing of frozen representations [7] is a classical proxy for representation quality. We include 22 linear probing tasks across diverse visual domains, using few-shot logistic regression to assess what information is encoded in model embeddings.

3. The MIEB Benchmark

MIEB provides a standardized, reproducible evaluation framework for image and image-text embedding models. We follow the MTEB [25] philosophy: fix model weights, evaluate zero-shot (or via lightweight linear probes), and aggregate across diverse tasks to yield a holistic view of model capabilities.

3.1. Task Categories

MIEB organizes 130 tasks into 8 categories, as shown in Table 1 and illustrated in Figure 1. We describe each below.

Retrieval (45 tasks). Retrieval tasks cover the broadest range of modalities and languages in MIEB. Queries and documents may each be images, text, or interleaved image-text sequences, yielding modality combinations such as image-to-text, text-to-image, and interleaved-to-image. This category includes multilingual retrieval datasets spanning 38 languages, such as xFlickr&CO, XM3600 [33], and WIT [4]. Evaluation uses nDCG@10.

Document Understanding (10 tasks). Inspired by the VIDORE benchmark [12], document understanding tasks test a model's ability to retrieve relevant document pages containing charts, tables, figures, and dense text. These tasks stress OCR-like understanding and the ability to match textual queries against visually complex page images. Evaluation uses nDCG@5.

Linear Probing (22 tasks). Linear probing evaluates the semantic richness of frozen embeddings by training a logistic regression classifier on top of them. We use 16 shots per class and a maximum of 100 training iterations. Datasets span fine-grained recognition benchmarks including ImageNet [10], CIFAR-10/100 [19], Oxford Pets [29], Oxford Flowers [26], Food-101 [3], Stanford Cars [18], and STL-10 [9].

Zero-shot Classification (23 tasks). Zero-shot classification uses image-text matching: for each image, the model scores it against text prompts of the form *"a photo of [class]"* and selects the highest-scoring class. This directly evaluates cross-modal alignment for visual concepts.

Task category	Example abilities assessed	# Tasks	# Languages	Modalities
Retrieval	cross-modal/-lingual matching	45	38	i-i; i-t; t-i; it-i; it-t; i-it; t-it; it-it; i-t
Document Understanding (Retrieval)	OCR abilities	10	2	t-i; i-t; it-t
Linear Probing (Classification)	information encoded	22	1	i-i; i-i
Clustering	embedding space consistency	5	1	i-i
Zero-shot Classification	cross-modal matching	23	1	i-t; i-t
Compositionality Evaluation (PairClassification)	reasoning with confounders	7	1	i-t; t-i
Vision-centric QA (Retrieval)	counting, object detection	6	1	it-t; it-i
Visual STS	OCR abilities	9	12	i-i
MIEB	all	130	38	all
MIEB-lite	all	51	38	all

Table 1. **An overview of MIEB tasks.** In brackets behind task categories, we denote the task type implementation in the code, e.g., our document understanding tasks use our retrieval implementation. We denote the modalities involved in both sides of the evaluation (e.g., queries and documents in retrieval; images and labels in zero-shot classification) with i=image, t=text.

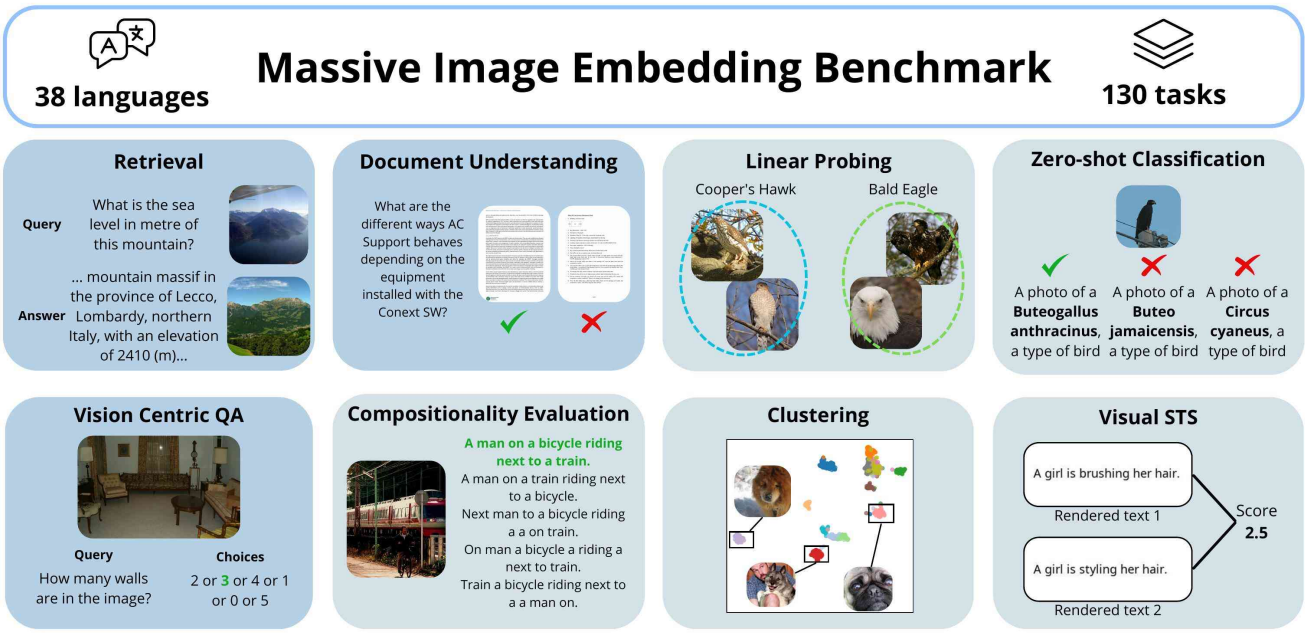

Figure 1. **Overview of MIEB task categories with examples.** MIEB covers 8 categories testing diverse vision and vision-language capabilities.

Clustering (5 tasks). Clustering evaluates embedding-space coherence by running k-means (with k equal to the number of ground-truth labels) on image embeddings and measuring Normalized Mutual Information (NMI) with ground-truth assignments. The UMAP visualization of image clusters for ImageNet Dog15 in Figure 4 illustrates the structured embedding space produced by strong encoders.

Compositionality (7 tasks). Compositionality tasks—ARO [36], WinoGround [34], and SugarCrepe [14]—test whether models are sensitive to fine-grained semantic distinctions, such as attribute-relation order, negation, and subtle visual differences. These are implemented as pair classification tasks and evaluated with accuracy.

Vision-centric QA (6 tasks). Vision-centric QA uses interleaved image-text queries (a question paired with an image context) to retrieve relevant image or text answers. These tasks probe capabilities like counting, object detection reasoning, and spatial understanding.

Visual STS (9 tasks). Visual STS is a novel contribution of MIEB. We take existing Semantic Textual Similarity benchmarks (STS12–STS17 [2] and STS-b [5]) and render each sentence as an image using a standard font. Models are then evaluated purely as image-image similarity systems using Spearman correlation with human judgments. This protocol isolates visual text comprehension: a model that has internalized textual semantics from image data will score

Model Name	# Params (M)	Runtime (NVIDIA H100 GPU hours)		
		MIEB	MIEB-lite	Reduction %
E5-V	8360	264.0	46.4	82.4% ↓
CLIP (base-patch32)	151	16.6	4.5	72.9% ↓

Table 2. **MIEB vs. MIEB-lite runtime comparison.**

well, while one that relies purely on visual features will not.

3.2. Interleaved Inputs

Several MIEB tasks involve interleaved image-text inputs, where queries or documents combine visual and textual content. For models that natively support interleaved inputs (e.g., MLLM-based encoders), these are processed directly. For encoder models lacking native interleaving support, we compute image and text embeddings separately and sum them to form a joint representation.

3.3. MIEB-lite

Running all 130 MIEB tasks can be expensive, especially for large MLLM-based models. For example, E5-V (8.36B parameters) requires 264 GPU hours (on NVIDIA H100s) to complete MIEB. We introduce **MIEB-lite**, a 51-task subset that preserves the diagnostic power of the full benchmark.

MIEB-lite is derived via task-correlation analysis: we compute per-task scores across all evaluated models, embed these score vectors with UMAP [24], and cluster them with HDBSCAN [23] to identify representative tasks from each cluster. We additionally exclude the most computationally expensive tasks while ensuring all 8 task categories and all supported languages are represented.

The result is a subset with strong full-benchmark correlation (Spearman $\rho = 0.992$) and 82.4% runtime reduction for large models, as shown in Table 2.

4. Models

We evaluate two families of image embedding models under a unified zero-shot protocol.

Contrastive Encoder Models. Encoder-based models are trained with contrastive objectives to align image and text representations. We evaluate CLIP variants [30] (including OpenCLIP [8] and DataComp [13] variants), SigLIP [37] (which replaces softmax contrastive with sigmoid binary cross-entropy), and EVA02-CLIP [31] (which leverages large-scale vision pretraining). These models are computationally efficient and excel at classification and clustering tasks.

MLLM-based Embedding Models. Recent work uses large multimodal language models as embedding engines. We evaluate E5-V [16], which instruction-tunes a multimodal LLM to produce universal embeddings by taking the last-token representation; VLM2Vec [17], which applies contrastive fine-tuning via LoRA [15] on Phi-3.5-Vision; and Voyage-multimodal-3 [1], a proprietary high-quality multimodal embedding model. MLLM-based models are substantially larger (billions of parameters) but offer richer semantic understanding, particularly for text-heavy visual content.

5. Implementation Details

All models are evaluated with frozen weights (zero-shot), except for linear probing where a logistic regression head is trained on frozen embeddings. We use the MTEB codebase [25] as the evaluation infrastructure, extended with vision task implementations.

Retrieval and Document Understanding. We use nDCG@10 for retrieval tasks and nDCG@5 for document understanding. For retrieval datasets with recall-focused annotations, we additionally report recall@1 and MAP@5. Corpus and query embeddings are computed with the same frozen model.

Clustering. We run k-means clustering with k equal to the number of ground-truth categories, repeated with 10 random seeds. Performance is measured by NMI between predicted and ground-truth cluster assignments.

Zero-shot Classification. Each image is matched against text embeddings of the form *"a photo of [class]"* via cosine similarity. Accuracy is reported.

Linear Probing. We extract embeddings for all examples, then train a logistic regression classifier using 16 labeled examples per class (max 100 iterations). Performance is measured by accuracy on the full test set. Figure 3 shows linear probing performance as a function of the number of shots k.

Compositionality. Pair classification accuracy is computed: given an image and two candidate text descriptions (or vice versa), the model must select the correct one based on embedding similarity.

Visual STS. Sentence pairs are rendered as 224×224 images using standard typography. Spearman correlation between predicted cosine similarities and human similarity ratings is computed.

MIEB-lite Derivation. Task-correlation vectors (per-task model scores) are projected with UMAP [24] into 2D and clustered with HDBSCAN [23]. One representative task per cluster is selected, prioritizing coverage of categories and languages while minimizing total evaluation cost.

6. Experimental Results

6.1. Main Results

Table 3 shows MIEB results for the top 20 models broken down by task category. The key finding is that *no single model dominates all categories*: different model families show complementary strengths.

Overall Rankings. Voyage-multimodal-3 achieves the highest overall mean score of 62.5 on MIEB Full (58.1 on MIEB-lite), driven by strong performance on Document Understanding (71.1) and Visual STS (81.8). Among encoder models, SigLIP-so400m-patch14-384 achieves the best encoder score of 57.1.

Encoder Models Excel at Classification and Clustering. Contrastive encoder models (SigLIP, CLIP, EVA02-CLIP) consistently outperform MLLM-based models on Linear Probing and Zero-shot Classification. EVA02-CLIP-bigE-14-plus achieves the highest LP score (86.0) and Clustering score (92.4). SigLIP-so400m-patch14-384 achieves the best ZS score (70.8) among all models. CLIP-ViT-bigG-14-laion2B achieves the highest Retrieval score (41.5). This reflects the discriminative nature of contrastive training at scale.

MLLM Models Excel on OCR and Multilingual Tasks. MLLM-based models (Voyage-multimodal-3, E5-V, VLM2Vec) shine on tasks requiring text comprehension within images and multilingual generalization. Voyage-multimodal-3 leads on Document Understanding (71.1) and Visual STS (81.8). E5-V achieves the best multilingual retrieval mean (74.3 across xFlickr&CO, XM3600, and WIT) and the highest Visual STS score among MLLM models (79.3). VLM2Vec variants achieve the best Vision-centric QA scores (62.0/62.1), outperforming all encoder models.

Compositionality Remains Challenging. Compositionality scores are consistently low across all models, with E5-V leading at 46.3 and most models scoring below 45. WinoGround and ARO reveal fundamental limitations in fine-grained compositional understanding that contrastive pretraining does not fully address.

6.2. Multilingual Retrieval

Table 4 shows performance on multilingual retrieval tasks across 38 languages. E5-V leads significantly with an average of 74.3 (variance 1.4), reflecting the advantage of large-scale multilingual LLM pretraining. SigLIP follows with 66.8 (variance 2.6), demonstrating that large-scale contrastive training with multilingual data also generalizes well. In contrast, VISTA [39] and VLM2Vec [17] show lower multilingual performance. OpenCLIP and EVA02-CLIP exhibit high variance across languages (7.3 and 7.2 respectively), indicating inconsistent multilingual coverage.

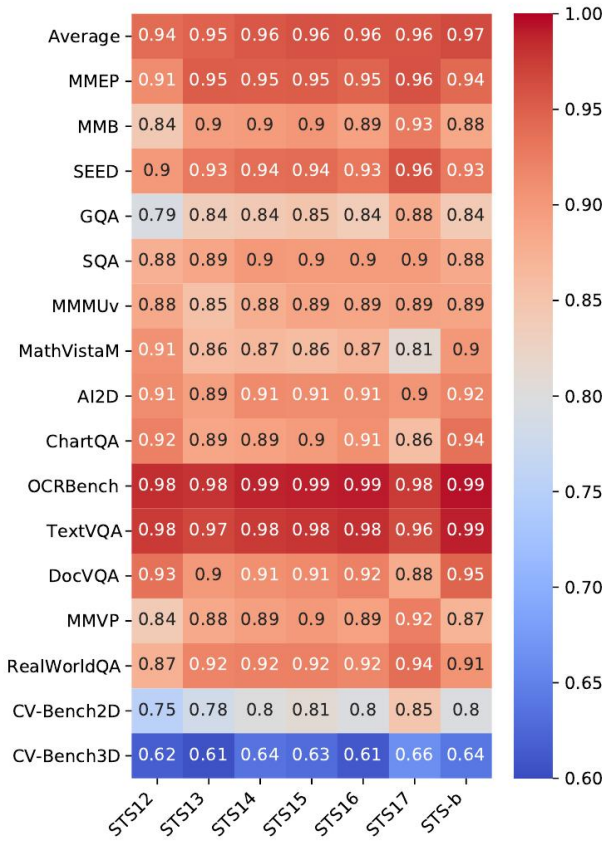

Figure 2. **Correlations between performance on generative MLLM benchmarks from Tong et al. [35] (y-axis) and our Visual STS (x-axis).** Visual STS is highly predictive of MLLM performance on OCR and chart understanding tasks.

6.3. Visual STS Analysis

Table 5 compares E5-V's performance on standard text STS and our Visual STS protocol. E5-V achieves an average of 86.0 on text STS but 79.3 on Visual STS, a gap of 6.7 points. This gap quantifies the remaining distance between text and visual text understanding. Notably, the gap varies across benchmarks: E5-V nearly matches text STS on STS17 (85.8 vs. 87.9) but shows larger gaps on STS12 (73.2 vs. 80.0).

Figure 2 shows that Visual STS scores are highly correlated ($> 99\%$) with performance on generative MLLM benchmarks [35] that require OCR and chart understand-

MIEB Full (130 tasks)													
Model Name (↓)	**Model Type**	**Rtrv. (45)**	**Clus. (5)**	**ZS. (23)**	**LP. (22)**	**Cmp. (7)**	**VC. (6)**	**Doc. (10)**	**vSTS (en) (7)**	**Rtrv. (m) (3 (55))**	**vSTS (x&m) (2 (19))**	**Mean (en) (125)**	**Mean (m) (130)**
Voyage-multimodal-3	MLLM	38.8	82.4	58.2	71.3	43.5	48.6	**71.1**	**81.8**	58.9	**70.4**	**62.0**	**62.5**
E5-V	MLLM	34.0	70.0	50.0	74.5	**46.3**	51.9	62.7	79.3	**66.6**	46.3	58.6	58.2
siglip-so400m-patch14-384	Enc.	40.8	82.1	**70.8**	84.6	40.4	46.3	56.4	68.0	40.2	41.4	61.2	57.1
siglip-large-patch16-384	Enc.	39.9	79.9	68.0	83.7	39.7	45.4	53.3	69.5	51.1	39.8	59.9	57.0
siglip-large-patch16-256	Enc.	38.8	82.1	67.7	82.5	40.8	44.9	39.4	67.4	49.8	38.1	57.9	55.2
siglip-base-patch16-512	Enc.	38.1	74.7	64.1	80.9	37.5	53.2	52.1	67.7	43.2	38.1	58.5	54.9
CLIP-ViT-bigG-14-laion2B	Enc.	**41.5**	85.6	69.4	83.6	42.4	43.2	43.2	70.9	28.0	34.5	60.0	54.2
siglip-base-patch16-384	Enc.	37.7	76.3	64.1	80.6	38.5	52.8	45.0	67.0	42.5	37.5	57.8	54.2
EVA02-CLIP-bigE-14-plus	Enc.	40.1	**92.4**	70.8	**86.0**	45.7	39.4	32.3	72.0	27.8	28.2	59.8	53.5
CLIP-ViT-L-14-DataComp.XL	Enc.	38.1	86.4	68.4	82.0	39.1	52.3	38.6	69.9	23.8	35.8	59.4	53.4
siglip-base-patch16-256(m)	Enc.	35.6	74.6	61.2	78.9	38.1	51.3	26.4	65.5	59.2	40.3	53.9	53.1
CLIP-ViT-H-14-laion2B	Enc.	39.7	83.9	67.5	82.5	42.0	45.8	40.4	65.5	25.5	33.9	58.4	52.7
CLIP-ViT-g-14-laion2B	Enc.	39.8	82.7	67.9	82.8	41.9	44.2	37.6	69.1	25.9	31.7	58.3	52.4
EVA02-CLIP-bigE-14	Enc.	39.0	89.4	69.3	84.5	42.4	43.6	31.6	68.8	25.5	28.3	58.6	52.2
siglip-base-patch16-256	Enc.	36.6	75.2	63.1	79.7	39.5	52.2	31.7	66.2	41.3	34.4	55.5	52.0
siglip-base-patch16-224	Enc.	36.3	74.5	62.6	79.3	39.8	51.1	26.2	64.3	41.2	33.5	54.3	50.9
CLIP-ViT-L-14-laion2B	Enc.	38.0	83.5	65.8	81.2	40.8	45.9	36.3	65.8	23.0	26.0	57.2	50.6
VLM2Vec-LoRA	MLLM	27.7	72.6	46.3	62.0	34.6	62.0	49.7	72.6	34.9	42.2	53.4	50.5
VLM2Vec-Full	MLLM	27.6	70.7	46.3	62.0	35.4	**62.1**	49.8	72.6	35.0	42.2	53.3	50.4
clip-vit-large-patch14	Enc.	33.7	76.4	62.1	80.1	44.8	44.1	38.0	64.5	20.2	35.1	55.4	49.9
MIEB-lite (51 tasks)													
Model Name (↓)	**Model Type**	**Rtrv. (11)**	**Clus. (2)**	**ZS. (7)**	**LP. (8)**	**Cmp. (6)**	**VC. (5)**	**Doc. (6)**	**vSTS (en) (2)**	**Rtrv. (m) (2 (47))**	**vSTS (x&m) (2 (19))**	**Mean (en) (47)**	**Mean (m) (51)**
Voyage-multimodal-3	MLLM	33.2	76.6	48.6	69.3	35.8	50.0	**63.5**	**84.2**	49.0	**70.4**	**57.7**	**58.1**
siglip-so400m-patch14-384	Enc.	32.4	75.9	73.8	78.8	32.8	48.0	46.9	69.6	35.4	41.4	57.3	53.5
siglip-large-patch16-384	Enc.	31.9	75.2	71.3	77.7	32.1	46.8	44.9	69.6	43.5	39.8	56.2	53.3
E5-V	MLLM	26.9	51.7	36.2	70.6	**39.4**	52.6	56.0	81.2	**58.3**	46.3	51.8	51.9
siglip-large-patch16-256	Enc.	31.0	76.5	70.3	76.3	33.4	46.5	31.9	67.6	42.6	38.1	54.2	51.4
CLIP-ViT-bigG-14-laion2B	Enc.	34.2	80.8	72.4	77.8	35.0	43.0	35.5	73.4	26.2	34.5	56.5	51.3
siglip-base-patch16-512	Enc.	30.8	69.7	66.3	74.6	29.7	55.5	42.6	67.1	34.8	38.1	54.5	50.9
EVA02-CLIP-bigE-14-plus	Enc.	**35.2**	**87.3**	**74.0**	**80.0**	38.9	38.8	26.2	73.7	26.0	28.2	56.8	50.8
siglip-base-patch16-384	Enc.	30.6	72.2	66.0	74.4	31.0	55.1	37.1	66.9	34.5	37.5	54.1	50.5
CLIP-ViT-L-14-DataComp.XL	Enc.	31.0	80.4	69.4	75.3	31.6	54.9	30.8	72.5	22.6	35.8	55.7	50.4
CLIP-ViT-H-14-laion2B	Enc.	32.8	79.3	69.4	76.8	34.8	46.8	33.7	68.3	23.9	33.9	55.2	50.0
EVA02-CLIP-bigE-14	Enc.	34.3	86.7	73.0	78.3	35.1	44.4	25.1	69.9	23.9	28.3	55.9	49.9
siglip-base-patch16-256(m)	Enc.	28.2	68.2	63.2	73.4	30.7	53.3	22.9	63.7	52.9	40.3	50.4	49.7
CLIP-ViT-g-14-laion2B	Enc.	33.5	76.8	69.6	77.3	34.7	45.0	29.9	71.6	24.2	31.7	54.8	49.4
siglip-base-patch16-256	Enc.	29.5	69.6	65.6	73.6	32.2	54.4	25.0	66.1	33.5	34.4	52.0	48.4
CLIP-ViT-L-14-laion2B	Enc.	31.1	76.4	67.8	75.9	33.6	46.9	28.7	68.7	21.4	26.0	53.6	47.6
clip-vit-large-patch14	Enc.	26.7	71.3	63.8	74.5	**39.4**	44.9	29.4	69.4	19.8	35.1	52.4	47.4
siglip-base-patch16-224	Enc.	29.3	68.4	65.0	73.5	32.5	53.0	20.9	64.2	33.6	33.5	50.8	47.4
CLIP-ViT-B-16-DataComp.XL	Enc.	28.3	73.6	61.9	73.2	31.4	56.9	22.7	69.7	19.9	28.5	52.2	46.6
VLM2Vec-LoRA	MLLM	21.0	66.3	32.1	64.8	29.4	**65.3**	42.7	70.9	24.8	42.2	49.1	46.0

Table 3. **MIEB results broken down by task categories for the top 20 models.** We provide averages of both English and multilingual tasks. Models are ranked by the Mean (m) column. Shortcuts are x=Crosslingual, m=Multilingual, en=English, and task categories from Figure 1. We refer to the leaderboard for the latest version: https://hf.co/spaces/mteb/leaderboard

Model Name	**xFlickr&CO**		**XM3600**		**WIT**		**avg.**	
	avg.	var.	avg.	var.	avg.	var.	avg.	var.
E5-V	**90.8**	**0.1**	**74.8**	3.5	**57.3**	0.6	**74.3**	1.4
SigLIP	80.4	1.2	65.6	5.3	54.4	1.3	66.8	2.6
VISTA (m3)	65.3	0.2	48.5	**2.0**	49.3	**0.4**	54.4	**0.9**
VLM2Vec	63.8	3.8	27.0	4.7	31.7	2.5	40.8	3.6
Open-CLIP	35.9	9.3	20.5	6.0	37.8	6.5	31.4	7.3
EVA02-CLIP	35.6	9.4	20.1	6.0	37.4	6.4	31.0	7.2

Table 4. **Performance of models on multilingual retrieval tasks across 38 languages.** We compute the average performance across languages (avg) and the respective variance (var). We take the best variant from each top-6 model family.

	12	**13**	**14**	**15**	**16**	**17**	**b**	**avg.**
STS*	80.0	89.9	85.7	89.1	85.9	87.9	83.5	86.0
v-STS (ours)	73.2	78.2	74.9	84.2	79.5	85.8	79.4	79.3

Table 5. **E5-V performance on regular STS and our Visual STS.** *: numbers from Jiang et al. [16]. Columns are STS12-17 and STS-b.

ing. This establishes Visual STS as a practical proxy metric: practitioners can select vision encoders for MLLM systems by evaluating Visual STS alone, without running full gener-

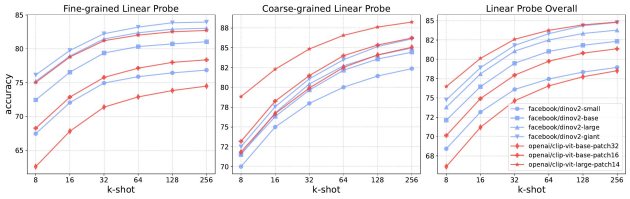

Figure 3. **Linear probing performance across different shots k.** Encoder models consistently outperform MLLM-based models across all shot counts, with performance plateauing around 16–32 shots.

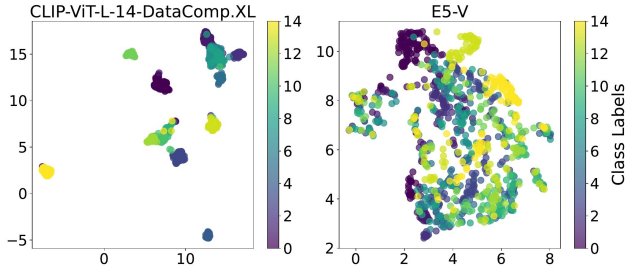

Figure 4. **UMAP Visualization of ImageNet Dog15.** Strong encoder models produce well-separated clusters corresponding to dog breeds, reflecting high NMI scores.

ative benchmarks.

6.4. Linear Probing: Effect of Shot Count

Figure 3 shows how linear probing accuracy varies with the number of labeled examples per class. Encoder models (SigLIP, EVA02-CLIP) consistently outperform MLLM-based models across all shot counts. Performance gains diminish after approximately 16–32 shots, justifying our choice of 16-shot evaluation as the primary protocol.

6.5. Clustering Visualization

Figure 4 shows a UMAP visualization of embeddings from ImageNet Dog15. Strong encoder models (EVA02-CLIP, SigLIP) produce tightly clustered, well-separated representations corresponding to individual dog breeds, consistent with their high NMI scores.

6.6. MIEB-lite Efficiency

Table 2 confirms that MIEB-lite dramatically reduces evaluation cost. For E5-V (8.36B parameters), runtime drops from 264 GPU hours to 46.4 GPU hours (82.4% reduction). For CLIP (151M parameters), runtime drops from 16.6 to 4.5 GPU hours (72.9% reduction). Combined with the near-perfect Spearman $\rho = 0.992$ correlation with full-benchmark rankings, MIEB-lite provides a practical path for rapid model development cycles.

7. Discussion

Complementary Model Families. Our results demonstrate that contrastive encoder models and MLLM-based embedding models occupy complementary niches. Encoder models dominate classification and clustering tasks that require discriminative visual features and broad visual concept coverage. MLLM-based models dominate tasks requiring semantic text understanding, multilingual generalization, and document comprehension. Practitioners should select models based on their specific use case rather than a single aggregate score.

Visual STS as a Proxy for MLLM Vision Encoder Selection. The strong correlation between Visual STS and MLLM OCR/chart benchmark performance has practical implications. When designing multimodal LLMs, practitioners often need to choose a vision encoder without running expensive generative evaluations. Our results suggest that Visual STS—which takes only minutes to evaluate—serves as a reliable proxy, enabling rapid encoder selection.

Interleaved Inputs. Tasks involving interleaved image-text inputs remain challenging. The simple additive fusion strategy (sum of image and text embeddings) used for encoder models is suboptimal, yet it is the only available option for models not natively supporting interleaving. Future work should develop better fusion strategies and train encoder models with interleaved inputs.

Compositional Reasoning Gap. Despite strong overall scores, all evaluated models struggle with compositional reasoning tasks (WinoGround, ARO, SugarCrepe). This persistent gap suggests that standard contrastive pretraining objectives do not incentivize fine-grained relational understanding. Closing this gap likely requires explicit compositional training data or architectural innovations.

8. Related Work

Vision-Language Pretraining. CLIP [30] established contrastive image-text pretraining as the dominant paradigm for learning general-purpose visual representations. SigLIP [37] improved training stability with a sigmoid loss formulation. EVA02-CLIP [31] scales to massive model sizes with strong performance. DataComp [13] contributed insights into data curation for contrastive pretraining. BLIP [21] and BLIP-2 [22] introduced bootstrapped captioning for improved vision-language alignment.

Text Embedding Benchmarks. MTEB [25] provided the unifying benchmark for text embeddings that inspired

MIEB. BEIR [32] standardized text retrieval evaluation. MMTEB [11] extended MTEB to multilingual text settings. Our work extends this unification paradigm to image and image-text embeddings.

Multimodal Embedding Models. VLM2Vec [17] and E5-V [16] demonstrated that instruction-tuned MLLMs can produce high-quality universal embeddings. VISTA [39] and BGE [6] contributed further architectures for multimodal retrieval. GME [38] and Nomic Embed [27] provide additional competitive baselines. Cambrian [35] provided MLLM benchmarks used as our reference for generative model evaluation.

Vision Benchmarks. Standard computer vision benchmarks including ImageNet [10], CIFAR [19], Oxford Pets [29], Oxford Flowers [26], Food-101 [3], and Stanford Cars [18] form the backbone of our linear probing and zero-shot classification evaluation. Compositionality benchmarks ARO [36], WinoGround [34], and Sugar-Crepe [14] probe relational reasoning. DINOv2 [28] provides self-supervised baselines.

9. Conclusion

We introduced MIEB, the Massive Image Embedding Benchmark, providing the most comprehensive and reproducible evaluation to date for image and image-text embeddings. Spanning 130 tasks, 8 categories, and 38 languages, MIEB reveals complementary strengths across model families: contrastive encoders excel at classification and clustering, while MLLM-based models excel at OCR, multilingual retrieval, and document understanding. Our novel Visual STS protocol establishes a strong proxy metric for MLLM vision encoder selection. MIEB-lite offers a practical 51-task subset with 82.4% runtime reduction for model development cycles. We release MIEB with a public leaderboard to standardize image embedding evaluation and guide future model development.

References

[1] voyage-multimodal-3: all-in-one embedding model for interleaved text, images, and screenshots. https://blog.voyageai.com/2024/11/12/voyage-multimodal-3/. 1, 2, 4

[2] Eneko Agirre, Daniel Cer, Mona Diab, Aitor Gonzalez-Agirre, and Weiwei Guo. *SEM 2013 shared task: Semantic textual similarity. In *Second Joint Conference on Lexical and Computational Semantics (*SEM), Volume 1: Proceedings of the Main Conference and the Shared Task: Semantic Textual Similarity*, pages 32–43, Atlanta, Georgia, USA, 2013. Association for Computational Linguistics. 3

[3] Lukas Bossard, Matthieu Guillaumin, and Luc Van Gool. Food-101 – mining discriminative components with random forests. In *European Conference on Computer Vision*, 2014. 2, 8

[4] Emanuele Bugliarello, Fangyu Liu, Jonas Pfeiffer, Siva Reddy, Desmond Elliott, Edoardo Maria Ponti, and Ivan Vulić. IGLUE: A benchmark for transfer learning across modalities, tasks, and languages. In *Proceedings of the 39th International Conference on Machine Learning*, pages 2370–2392. PMLR, 2022. 2

[5] Daniel Cer, Mona Diab, Eneko Agirre, Iñigo Lopez-Gazpio, and Lucia Specia. SemEval-2017 task 1: Semantic textual similarity multilingual and crosslingual focused evaluation. In *Proceedings of the 11th International Workshop on Semantic Evaluation (SemEval-2017)*, pages 1–14, Vancouver, Canada, 2017. Association for Computational Linguistics. 3

[6] Jianlv Chen, Shitao Xiao, Peitian Zhang, Kun Luo, Defu Lian, and Zheng Liu. Bge m3-embedding: Multi-lingual, multi-functionality, multi-granularity text embeddings through self-knowledge distillation. *arXiv preprint arXiv:2402.03216*, 2024. 8

[7] Xinlei Chen, Saining Xie, and Kaiming He. An empirical study of training self-supervised vision transformers. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 9640–9649, 2021. 2

[8] Mehdi Cherti, Romain Beaumont, Ross Wightman, Mitchell Wortsman, Gabriel Ilharco, Cade Gordon, Christoph Schuhmann, Ludwig Schmidt, and Jenia Jitsev. Reproducible scaling laws for contrastive language-image learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 2818–2829, 2023. 4

[9] Adam Coates, Andrew Ng, and Honglak Lee. An analysis of single-layer networks in unsupervised feature learning. In *Proceedings of the Fourteenth International Conference on Artificial Intelligence and Statistics*, pages 215–223, Fort Lauderdale, FL, USA, 2011. PMLR. 2

[10] Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. Imagenet: A large-scale hierarchical image database. In *2009 IEEE conference on computer vision and pattern recognition*, pages 248–255. Ieee, 2009. 2, 8

[11] Kenneth Enevoldsen, Isaac Chung, Imene Kerboua, Márton Kardos, Ashwin Mathur, David Stap, Jay Gala, Wissam Siblini, Dominik Krzemiński, Genta Indra Winata, Saba Sturua, Saiteja Utpala, Mathieu Ciancone, Marion Schaeffer, Gabriel Sequeira, Diganta Misra, Shreeya Dhakal, Jonathan Rystrøm, Roman Solomatin, Ömer Çağatan, Akash Kundu, Mar-

tin Bernstorff, Shitao Xiao, Akshita Sukhlecha, Bhavish Pahwa, Rafał Poświata, Kranthi Kiran GV, Shawon Ashraf, Daniel Auras, Björn Plüster, Jan Philipp Harries, Loïc Magne, Isabelle Mohr, Mariya Hendriksen, Dawei Zhu, Hippolyte Gisserot-Boukhlef, Tom Aarsen, Jan Kostkan, Konrad Wojtasik, Taemin Lee, Marek Šuppa, Crystina Zhang, Roberta Rocca, Mohammed Hamdy, Andrianos Michail, John Yang, Manuel Faysse, Aleksei Vatolin, Nandan Thakur, Manan Dey, Dipam Vasani, Pranjal Chitale, Simone Tedeschi, Nguyen Tai, Artem Snegirev, Michael Günther, Mengzhou Xia, Weijia Shi, Xing Han Lù, Jordan Clive, Gayatri Krishnakumar, Anna Maksimova, Silvan Wehrli, Maria Tikhonova, Henil Panchal, Aleksandr Abramov, Malte Ostendorff, Zheng Liu, Simon Clematide, Lester James Miranda, Alena Fenogenova, Guangyu Song, Ruqiya Bin Safi, Wen-Ding Li, Alessia Borghini, Federico Cassano, Hongjin Su, Jimmy Lin, Howard Yen, Lasse Hansen, Sara Hooker, Chenghao Xiao, Vaibhav Adlakha, Orion Weller, Siva Reddy, and Niklas Muennighoff. Mmteb: Massive multilingual text embedding benchmark, 2025. 2, 8

[12] Manuel Faysse, Hugues Sibille, Tony Wu, Bilel Omrani, Gautier Viaud, Céline Hudelot, and Pierre Colombo. Colpali: Efficient document retrieval with vision language models, 2024. 2

[13] Samir Yitzhak Gadre, Gabriel Ilharco, Alex Fang, Jonathan Hayase, Georgios Smyrnis, Thao Nguyen, Ryan Marten, Mitchell Wortsman, Dhruba Ghosh, Jieyu Zhang, et al. Datacomp: In search of the next generation of multimodal datasets. *Advances in Neural Information Processing Systems*, 36, 2024. 4, 7

[14] Cheng-Yu Hsieh, Jieyu Zhang, Zixian Ma, Aniruddha Kembhavi, and Ranjay Krishna. Sugarcrepe: Fixing hackable benchmarks for vision-language compositionality. In *Thirty-Seventh Conference on Neural Information Processing Systems Datasets and Benchmarks Track*, 2023. 2, 3, 8

[15] Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, Weizhu Chen, et al. Lora: Low-rank adaptation of large language models. *ICLR*, 1(2):3, 2022. 4

[16] Ting Jiang, Minghui Song, Zihan Zhang, Haizhen Huang, Weiwei Deng, Feng Sun, Qi Zhang, Deqing Wang, and Fuzhen Zhuang. E5-v: Universal embeddings with multimodal large language models. *arXiv preprint arXiv:2407.12580*, 2024. 1, 2, 4, 6, 8

[17] Ziyan Jiang, Rui Meng, Xinyi Yang, Semih Yavuz, Yingbo Zhou, and Wenhu Chen. Vlm2vec: Training vision-language models for massive multimodal embedding tasks. *arXiv preprint arXiv:2410.05160*, 2024. 1, 2, 4, 5, 8

[18] Jonathan Krause, Jia Deng, Michael Stark, and Li Fei-Fei. Collecting a large-scale dataset of fine-grained cars. 2013. 2, 8

[19] Alex Krizhevsky, Vinod Nair, and Geoffrey Hinton. Cifar-10 (canadian institute for advanced research). 2, 8

[20] LAION-AI. Clip benchmark. https://github.com/LAION-AI/CLIP_benchmark, 2024. Accessed: 2024-11-11. 2

[21] Junnan Li, Dongxu Li, Caiming Xiong, and Steven Hoi. Blip: Bootstrapping language-image pre-training for unified vision-language understanding and generation. In *International conference on machine learning*, pages 12888–12900. PMLR, 2022. 7

[22] Junnan Li, Dongxu Li, Silvio Savarese, and Steven Hoi. Blip-2: bootstrapping language-image pre-training with frozen image encoders and large language models. In *Proceedings of the 40th International Conference on Machine Learning*. JMLR.org, 2023. 7

[23] Leland McInnes, John Healy, Steve Astels, et al. hdbscan: Hierarchical density based clustering. *J. Open Source Softw.*, 2(11):205, 2017. 1, 4, 5

[24] Leland McInnes, John Healy, and James Melville. Umap: Uniform manifold approximation and projection for dimension reduction. *arXiv preprint arXiv:1802.03426*, 2018. 1, 4, 5

[25] Niklas Muennighoff, Nouamane Tazi, Loic Magne, and Nils Reimers. MTEB: Massive text embedding benchmark. In *Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics*, pages 2014–2037, 2023. 1, 2, 4, 7

[26] Maria-Elena Nilsback and Andrew Zisserman. Automated flower classification over a large number of classes. In *2008 Sixth Indian Conference on Computer Vision, Graphics & Image Processing*, pages 722–729, 2008. 2, 8

[27] Zach Nussbaum, John X. Morris, Brandon Duderstadt, and Andriy Mulyar. Nomic embed: Training a reproducible long context text embedder, 2024. 8

[28] Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, Francisco Massa, Alaaeldin El-Nouby, et al. Dinov2: Learning robust visual features without supervision. *Transactions on Machine Learning Research Journal*, pages 1–31, 2024. 8

[29] Omkar M Parkhi, Andrea Vedaldi, Andrew Zisserman, and C. V. Jawahar. Cats and dogs. In *2012 IEEE Conference on Computer Vision and Pattern Recognition*, pages 3498–3505, 2012. 2, 8

[30] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sas-

try, Amanda Askell, Pamela Mishkin, Jack Clark, et al. Learning transferable visual models from natural language supervision. In *International conference on machine learning*, pages 8748–8763. PMLR, 2021. 1, 2, 4, 7

[31] Quan Sun, Yuxin Fang, Ledell Wu, Xinlong Wang, and Yue Cao. Eva-clip: Improved training techniques for clip at scale. *arXiv preprint arXiv:2303.15389*, 2023. 1, 4, 7

[32] Nandan Thakur, Nils Reimers, Andreas Rücklé, Abhishek Srivastava, and Iryna Gurevych. Beir: A heterogeneous benchmark for zero-shot evaluation of information retrieval models. In *Thirty-fifth Conference on Neural Information Processing Systems Datasets and Benchmarks Track (Round 2)*, 2021. 2, 8

[33] Ashish V Thapliyal, Jordi Pont Tuset, Xi Chen, and Radu Soricut. Crossmodal-3600: A massively multilingual multimodal evaluation dataset. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pages 715–729, 2022. 2

[34] Tristan Thrush, Ryan Jiang, Max Bartolo, Amanpreet Singh, Adina Williams, Douwe Kiela, and Candace Ross. Winoground: Probing vision and language models for visio-linguistic compositionality. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 5238–5248, 2022. 2, 3, 8

[35] Shengbang Tong, Ellis Brown, Penghao Wu, Sanghyun Woo, Manoj Middepogu, Sai Charitha Akula, Jihan Yang, Shusheng Yang, Adithya Iyer, Xichen Pan, et al. Cambrian-1: A fully open, vision-centric exploration of multimodal llms. *arXiv preprint arXiv:2406.16860*, 2024. 5, 8

[36] Mert Yuksekgonul, Federico Bianchi, Pratyusha Kalluri, Dan Jurafsky, and James Zou. When and why vision-language models behave like bags-of-words, and what to do about it? In *International Conference on Learning Representations*, 2023. 2, 3, 8

[37] Xiaohua Zhai, Basil Mustafa, Alexander Kolesnikov, and Lucas Beyer. Sigmoid loss for language image pre-training. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 11975–11986, 2023. 1, 2, 4, 7

[38] Xin Zhang, Yanzhao Zhang, Wen Xie, Mingxin Li, Ziqi Dai, Dingkun Long, Pengjun Xie, Meishan Zhang, Wenjie Li, and Min Zhang. Gme: Improving universal multimodal retrieval by multimodal llms, 2024. 8

[39] Junjie Zhou, Zheng Liu, Shitao Xiao, Bo Zhao, and Yongping Xiong. Vista: Visualized text embedding for universal multi-modal retrieval. *arXiv preprint arXiv:2406.04292*, 2024. 2, 5, 8